

Prevention and Follow up of Bacterial Meningitis-related Hearing Loss

Introduction:

Dexamethasone is beneficial for treatment in *H.Infleunza* and *S. pneumoniae* meningitis to diminish the risk of hearing loss.

Hearing deficit is a major complication of bacterial meningitis. It may be present at admission or may develop during the course of the disease. Bacteria most likely to cause hearing loss are *S.pneumonia* (32%) and *H.Infleunza*. Less cases of *H.Infleunza* meningitis have been reported mostly owing to the introduction of the conjugate vaccine.

Incidence of bacterial meningitis related hearing loss reached 35%. A review of the literature showed an incidence of permanent hearing loss of 10% and an incidence of 4% for profound bilateral hearing loss. There is a risk of cochlear ossifications as early as 2-4 weeks after recovery from meningitis. Formal audiology assessment should be performed as soon as possible after diagnosis of meningitis for all children.

Recently, we noticed an increase of cases presented with *H.influenza* complicated meningitis. Therefore, a modification in the management of meningitis is warranted.

Objectives:

To provide a clear guideline for prevention and follow up of bacterial meningitis related hearing loss.

I. Adjunctive Steroid Therapy

In a Cochrane meta-analysis, corticosteroids were found to decrease overall hearing loss and neurological sequelae. No excess of dexamethasone -related adverse effects was observed compared to placebo group. A subgroup analysis showed that corticosteroids reduced mortality in pneumococcal meningitis but not in other pathogen related meningitis.

❖ **Patients eligible for dexamethasone:**

1. Infants more than 8 weeks of age AND
2. Clinically suspected meningitis with CSF findings suggestive of bacterial meningitis

Note: If CSF sampling is not done, then do not give dexamethasone unless advised by ID team in highly suspicion of bacterial meningitis

❖ **CSF findings suggestive of bacterial meningitis**

- I. **CSF pleocytosis of mainly polymorphic leukocytes AND low glucose concentration (low CSF to blood glucose ratio) AND /OR** elevated protein level.
- II. AND/OR cloudy cerebrospinal fluid (CSF)
- III. AND/OR positive gram stain in CSF

❖ **CSF sampling**

Sample should be sent for the following:

- I. Microscopy
- II. Biochemistry
- III. Culture
- IV. Bacterial antigen
- V. Bacterial PCR
- VI. Viral PCR

❖ **Timing of dexamethasone treatment:**

Before or with the first dose of antibiotics and can still be started up to 4 hours after initiation of antibiotic treatment.

❖ **Dose:**

Dexamethasone: 0.15 mg/kg/dose IV Q6H for 4 days.

❖ **Discontinuation of Dexamethasone:**

- If CSF result is not suggestive of bacterial meningitis OR
- If *H. influenzae* and *streptococcus pneumoniae* are not identified in CSF culture/PCR within 48 h.

II. **Audiological management for bacterial meningitis**

- Referral is the responsibility of the medical team caring for the child.
- This part of the protocol need to be in place to ensure referral from the pediatric wards or NICU/SCBU to Otolaryngology/ Audiology clinics.
- Otolaryngology/ Audiology teams are responsible for arranging appointments and follow-ups.

❖ **Target population:** Children from birth to age 12 years who have been treated for bacterial meningitis.

❖ **Recommendation:** Children with meningitis should be referred for audiological testing as soon as possible after diagnosis; they should be referred urgently to Otolaryngology/ Audiology clinic at regional hospital or Al Nahdha hospital for hearing assessment and follow up. An Outpatient appointment should be scheduled within 7-10 days.

❖ **Follow up and monitoring of hearing loss after meningitis at Otolaryngology/ Audiology clinic :**

- If initial hearing assessment showed normal hearing, they should be followed up at 1, 3, 6 and 12 months and annually thereafter until they are three years post meningitis
- Cases of mild to moderate hearing losses should be retested within 2 weeks, if they have stable result they should be tested audiological every 3 months for the first year; every 6 months after that, until they are 3 years post meningitis.
- Children who have fluctuating or worsening SNHL should be referred urgently to cochlear implant team (Al Nahdha hospital)

❖ **Follow up with Primary team.**

❖ **Note: if it is complicated meningitis an additional follow up with Infectious Diseases team is required.**

References:

1- D.van de Beek, C.Cabellos, O.Dzupova, et al. ESCMID guideline: Diagnosis and treatment of acute bacterial meningitis. Clin Microbiol Infect 2016; 22: S37-S62

2- Brouwer MC, McIntyre P, Prasad K, et al. Corticosteroids for acute Bacterial meningitis (Review). Cochrane Database of Systematic Reviews 2015